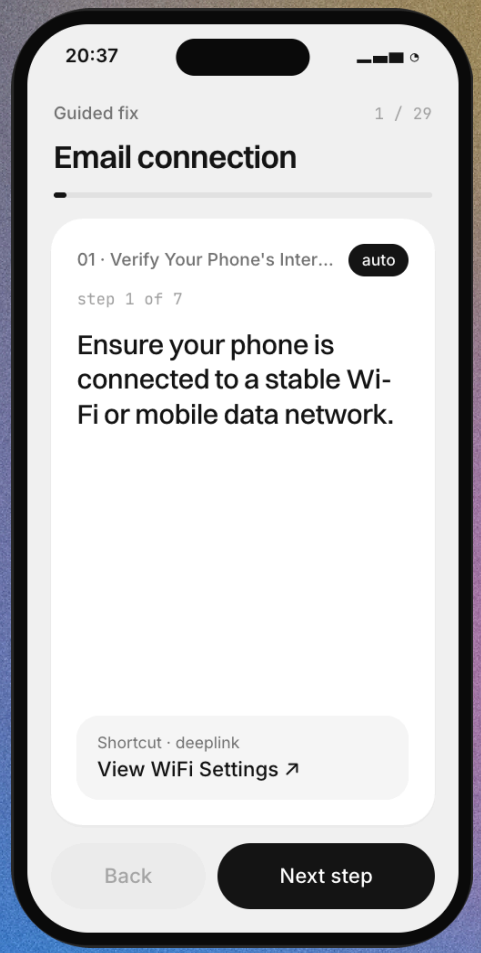
Smart Guided Troubleshooting Engine

Vague complaint in. Safe, ordered, one-tap fix plan out. Never invented.



Samsung PRISM · Gen AI Hackathon 3.0 · Theme 02

Heavenly Restriction — Aryan Singh · Rajat Pandey

The problem

Customers describe Galaxy problems vaguely. Today an agent reads a Samsung article and types the fix by hand: about 15 minutes a scenario, across millions of calls.

Translation

Vague words to the right article

“Screen flickers and the battery dies fast”

Structuring

Prose to ordered, atomic steps

Tap Display → Tap Screen timeout

The last mile

One tap, not a menu hunt

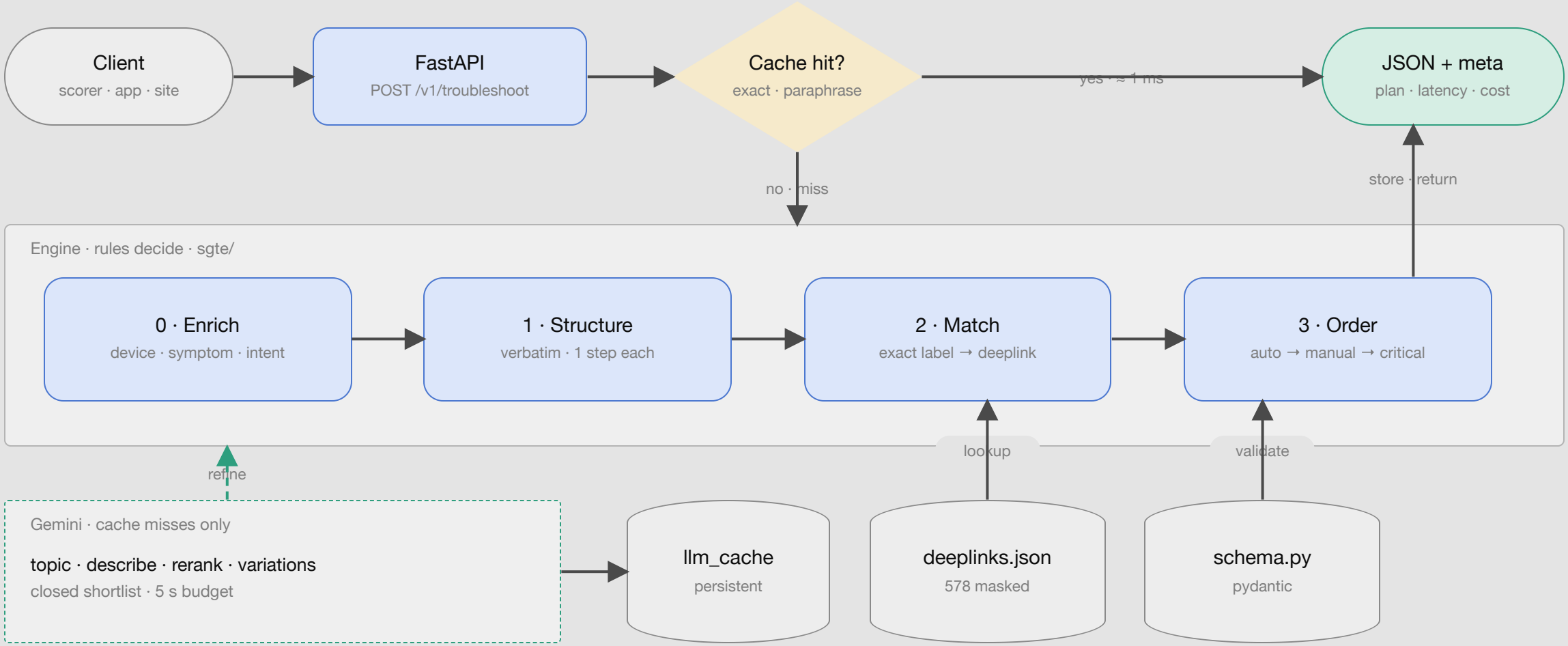
Deeplink opens the exact Settings screen

Caveats we designed around

Caveat	Design answer
LLMs invent steps and links	Steps copied from the article; links only from the catalogue
A close deeplink is a wrong one	Exact-label matching, never the parent menu
Not every fix lives in Settings	No link rather than a guess
Some steps are dangerous	Critical steps always come last
Not every article is a how-to	Empty plan, never invented steps
Wording varies, models are slow	Article-keyed cache, about 1 ms

Core architecture

Rules decide, Gemini refines.



Low-level design

Inside each module

Structure · parse.py · engine.py

Clean markup, drop URL lines → Sections and labelled groups → Imperative filter, one tap per step → Category, then order

Every step is a substring of the article · 349 / 349

Deeplink matcher · catalog.py

Extract target label → Exact-label index (578) → Polarity: open · on · off → BM25 + filters, else dummy / none

572 / 572 catalogue entries resolve exactly

Cache · cache.py

Exact key: article + query → Paraphrase: same article, keywords → No article: score ≥ 0.45 → Miss: engine, then store

p50: 0.12 ms exact · 1.09 ms paraphrase

Gemini wrapper · llm.py

Memo hit? → Call model (2.5 s each) → Validate JSON and rules → Accept, else rules' answer

Never writes steps or URIs · 5 s budget

Response contract

Goal → Actions → StepGroups → actionable + validation deeplinks

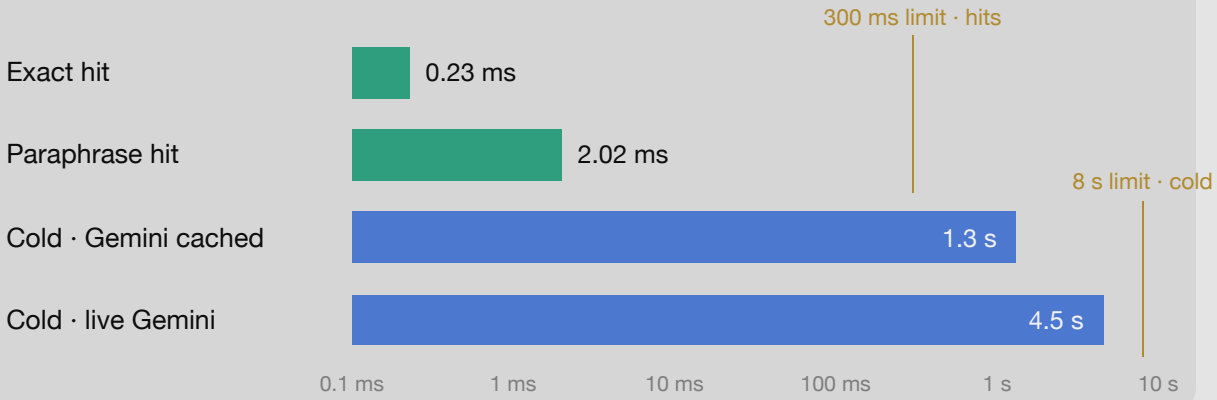
Results

Live demo samsung-prism-sentinel.onrender.com

Video youtu.be/S4uCkcCt1sY

Latency p95 against the limit

log scale · shorter is better



Exact-screen deeplinks

share of 572 catalogue entries resolved exactly



Retrieval alone lands on neighbouring settings.

Step accuracy 2.91 / 3



Check	Result	Limit
G2 · /health	ok	200
G3 · coverage	20 / 20	≥ 95%
G4 · schema-valid	20 / 20	≥ 90%
G5 · URL leaks	0	0
A3 · paraphrase hits	216 / 216	≥ 80%
A4 · unseen formats	14 / 14	valid

Built for Samsung

Agent assist

A plan in about 1 ms, not 15 minutes

Self-service

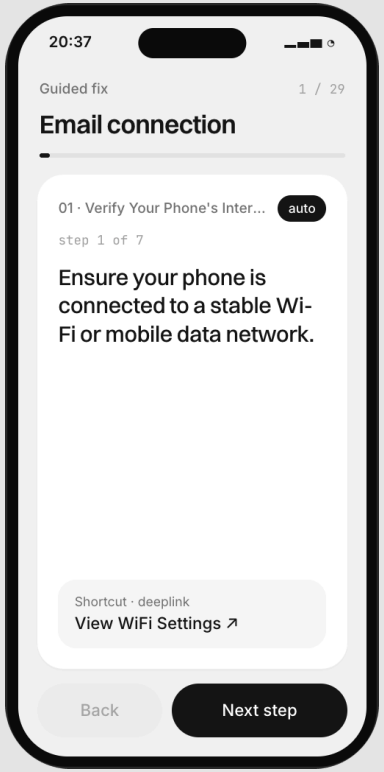
One tap to the exact screen

Voice & Bixby

Steps and validation drive the fix

After OTA updates

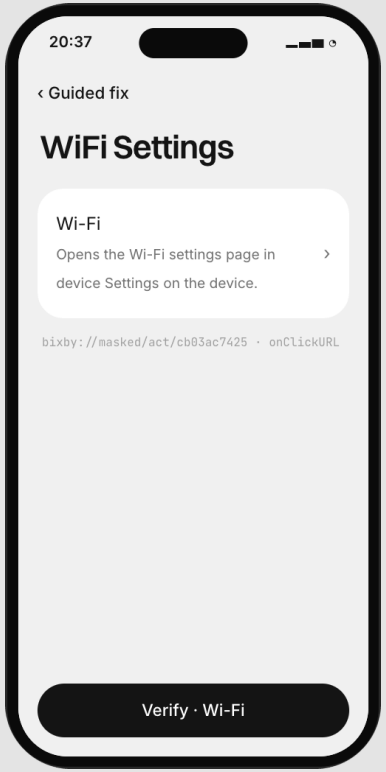
Same article, same answer



01

Guided steps

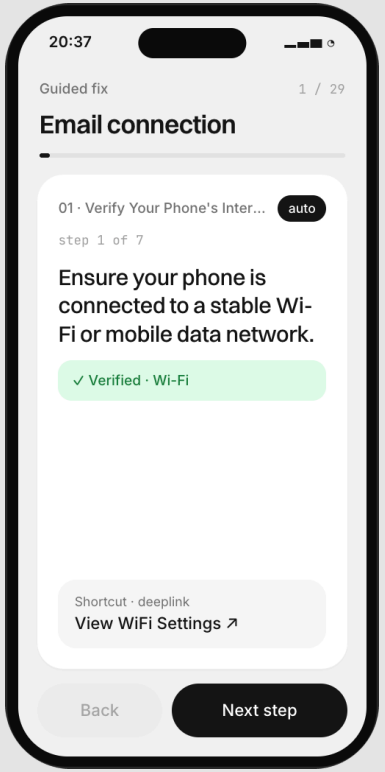
From the article, one interaction each



02

One tap to Settings

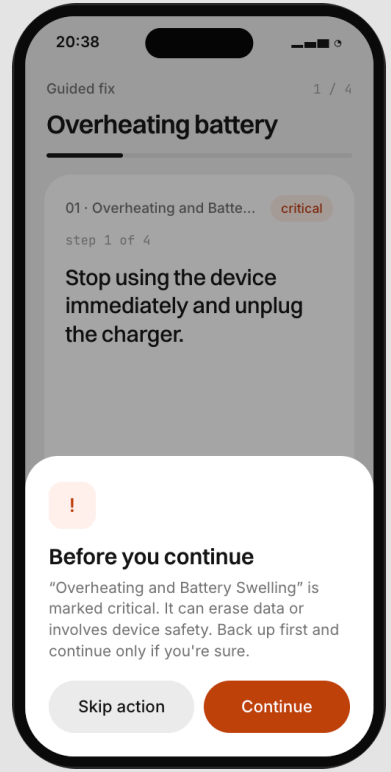
The deeplink opens the exact screen



03

Fix verified

Validation reads the setting back



04

Critical, confirmed

Safety steps always ask first

Screens captured from the live simulator at samsung-prism-sentinel.onrender.com

Scales by adding data, not code

The brief's production target is 10k+ scenarios.

Area	Today	At 10k+ scenarios
New scenario	Add an article, no code	Unchanged
New setting	Add a catalogue row	Unchanged
Cache	In memory, 2,048 articles	Redis shared store
Lookup	Scan of cached articles	ANN vector index
Embeddings	Hashed vectors + BM25	Sentence embeddings, same interface
Reach	English, one process	Hindi and Korean, more workers

Honest limits, and what is next

Limit	Why we accept it
20 of 83 kit step groups link	A wrong link is worse than none
Completeness capped at 0.91	The verbatim rule: no invented steps
Hashed vectors, in-memory cache	Swappable behind small interfaces
Scored against the articles	The kit has no ground truth

Next

Engineer's view with a decision trace
Live SIIS and outcome telemetry
Hindi and Korean queries
On-device refinement model

Thank you

Questions?

Live	samsung-prism-sentinel.onrender.com
Code	github.com/Blazehue/Samsung_PRISM-Sentinel
Release tag	PRISM_GENAI_HACKATHON_Y2026
Video	youtu.be/S4uCkcCt1sY

Visual style inspired by “Mobile Product Review” by Natasha Tenggoro (Figma Community, CC BY 4.0)